EVERY POINT OF THE IMAGE OF A SMOOTH MORPHISM ADMITS AN ETALE LOCAL SECTION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10845-10857 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 10859-10868, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Only the author's FIRST proof (lines 10859-10868) is retained; the chapter contains a second proof immediately below, outside the collected spans.

Editorial note. The statement's reference environment cites \cite[Corollaire 17.16.3 (ii)]{EGA4}; a bibliography.tex holding the snapshot my.bib EGA4 entry verbatim is attached (precedent S3-morph-bertini-irreducible, CONTROL section 8.1).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The lemma that smooth fibres over a field have dense closed points with finite separable residue fields (Varieties) and the slice lemma producing an etale locally closed slice through such a point (same chapter); cited inputs with the two-step argument written in the retained span.

Source reference. Original lemma 057G: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10775–10783. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 056U: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10868–10868. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Smooth morphisms admit étale local sections. See* [1, Corollaire 17.16.3 (ii)]. *Let $f : X \to S$ be a smooth morphism of schemes. Let $s \in S$ be a point in the image of f. Then there exists an étale neighbourhood $(S', s') \to (S, s)$ and a S-morphism $S' \to X$.*

First proof of Lemma 0.1. By assumption $X_s \neq \emptyset$. By Varieties, Lemma 056U there exists a closed point $x \in X_s$ such that $\kappa(x)$ is a finite separable field extension of $\kappa(s)$. Hence by Lemma 057G there exists an immersion $Z \to X$ such that $Z \to S$ is étale and such that $x \in Z$. Take $(S', s') = (Z, x)$. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

References

[1] Grothendieck, Alexander and Dieudonné, Jean. *Éléments de géométrie algébrique IV*. Publications Mathématiques 20, 24, 28, 32. Institute des Hautes Études Scientifiques, 1964–1967.